Group 25

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Table of Contents

01 Problem Statement

02 How The System Works

03 Circuit Wiring Overview

04 Arduino Code Walkthrough

05 TinkerCAD Simulation

06 Conclusion

Problem Statement

01 Manual Contact Risk

Traditional sanitizer dispensers require physical contact, increasing the risk of germ transmission.

02 Sanitizer Wastage

Manual dispensers often result in overuse or spillage, leading to unnecessary consumption and cost.

03 Need for Automation

Public spaces need efficient, touchless hygiene solutions that minimize human intervention and maximize safety.

04 Our Solution

A microprocessor-controlled automatic hand sanitizer dispenser using Arduino UNO, IR sensor, and smart controls.

System Components

Arduino UNO

Microprocessor / Brain

Reads sensor input, runs control logic, triggers relay and LED outputs.

Ultrasonic Sensor

Detection Unit

Detects hand presence within ~10–15 cm range. Outputs LOW when object detected.

Relay Module

Switching Circuit

Electrically switches the pump ON/OFF based on Arduino signal. Isolates pump from Arduino.

DC Water Pump

Actuator

Draws and dispenses sanitizer when activated. Powered by separate supply.

LED Indicator

Visual Feedback

Lights up green while sanitizer is being dispensed. Turns off after cycle completes.

Power Supply

Energy Source

Arduino via USB or 9V adapter. Pump via 4×AA batteries or 5V DC adapter.

How The System Works

1

Hand Approaches

User places hand under IR sensor

2

Sensor Detects

IR sensor outputs LOW signal to Arduino

3

Arduino Decides

Checks cooldown, triggers relay if clear

4

Pump Activates

Relay closes, pump runs for 1.5 seconds

5

LED Lights Up

Green LED ON during dispensing

6

Cycle Resets

Pump & LED off, cooldown begins (3s)

Cooldown of 3 seconds prevents continuous dispensing — ensuring efficiency and preventing waste

Circuit Wiring Overview

Pin Connection Table

Component	Pin	→	Arduino	Mode
Ultrasonic Sensor	VCC	→	5V	Power
Ultrasonic Sensor	GND	→	GND	Ground
Ultrasonic Sensor	OUT	→	Pin 9	INPUT
Relay	VCC	→	5V	Power
Relay	GND	→	GND	Ground
Relay	IN	→	Pin 10	OUTPUT
LED (+)	Anode	→	Pin *	OUTPUT
LED (-)	Cathode	→	GND	Ground
Pump (+)	via Relay	→	Ext. PSU	Switched

** LED connected via 220Ω resistor in series*

⚠ Important Notes

🔋 Separate Power

Always power the pump from an external supply — not from the Arduino 5V pin. The pump draws too much current.

🔀 Common Ground

Connect Arduino GND and external power supply GND together. Without this, the relay will not work.

⚡ Relay Type

Most relay modules are ACTIVE LOW. Sending LOW from Arduino Pin 8 turns the pump ON.

🔧 IR Sensitivity

Adjust the potentiometer on the IR sensor to set the detection distance to ~10–15 cm.

Arduino Code Walkthrough

The code

```
// Pin Setup (setup())
const int trigPin = 9;
const int echoPin = 10;
const int servoPin = 11;

Servo dispenserServo;
dispenserServo.attach(servoPin);

// Distance Calculation
long getDistance() {
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  long duration = pulseIn(echoPin, HIGH);
  return duration * 0.034 / 2;
}

// Main Logic (loop())
if (distance > 2 && distance < 10) {
  digitalWrite(greenLed, HIGH);
  dispenserServo.write(90);  // Push pump
  delay(1000);
  dispenserServo.write(0);   // Reset

  while (getDistance() < 15) {
    delay(100);  // Wait for hand removal
  }
```

Pin Setup (setup())

Configures Pins 9 (Trigger), 10 (Echo), 11 (Servo) as INPUT or OUTPUT. Initializes servo to retracted position (0°

Sensor Reading (loop())

Sends ultrasonic pulse from Pin 9. Measures echo time on Pin 10. Converts to distance in cm using speed of sound formula.

Cooldown Logic

Detects hand when distance is between 2-10 cm. Triggers dispenser and activates green LED

Dispense() Function

Servo rotates to 90° (pushes pump) for 1 second, then returns to 0°. Prevents re-dispensing until hand is removed (>15cm away).

TinkerCAD Simulation

01 Go to tinkercad.com

Create a free account. Click Circuits → Create New Circuit.

02 Add Arduino UNO

Search 'Arduino UNO' in the component panel. Drag to canvas.

03 Add PIR Sensor

TinkerCAD has no IR sensor. Use PIR Motion Sensor as substitute.
Wire: VCC→5V, GND→GND, OUT→Pin 7.

04 Add Transistor + Motor

Use NPN Transistor (2N2222) for relay and DC Motor for pump.
Base→Pin 8 (via 1kΩ), Collector→Motor(+), Emitter→GND.

05 Add LED

Add green LED. Anode → 220Ω resistor → Pin 13. Cathode → GND.

06 Paste Code & Simulate

Click Arduino → Code → Text. Paste smart_sanitizer.ino code. Click Start Simulation.

TinkerCAD substitutions: PIR Sensor = IR Sensor | NPN Transistor = Relay | DC Motor = Pump

Conclusion

Key Achievements

Successfully built contactless sanitizer system with Arduino

Challenges

Sensor calibration, relay timing, power management

What We Learned

Microprocessor integration, sensor-actuator control, circuit design

Testing Results

System detects hands accurately, dispenses consistently

Future Improvements

Add LCD display, sanitizer level sensor, battery backup

Impact

Promotes hygiene, reduces contamination in public spaces

Promoting hygiene through smart automation — contactless, efficient, and reliable